

Report of the Paragraph Classification of Jin Yong's Novels Based on LDA Topic Modeling

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Abstract

This is a Report of the Paragraph Classification of Jin Yong's Novels Based on LDA Topic Modeling. It employs the Latent Dirichlet Allocation (LDA) model in conjunction with a Support Vector Machine (SVM) classifier to conduct thematic modeling and classification experiments on novel paragraphs. By adjusting parameters such as the number of topics and paragraph length, the study systematically investigates the impact of various variables on classification performance. The experimental data indicate that : there exists an optimal range for the number of topics; regardless of whether words or characters are used as the basic units, there is a significant positive nonlinear correlation between paragraph length and mean classification accuracy; there are differences in classification performance between word units and character units (word units had a significant advantage in semantic expression, particularly in long text (K=3000) scenarios, where their peak accuracy exceeded that of character units.) The highest accuracy rate of 98.1% is achieved with a configuration based on basic units of words and 10 topics when the paragraph length reaches 3000 words.

Introduction

Text classification and topic modeling are fundamental tasks in natural language processing (NLP), with applications ranging from document organization to semantic analysis. In this study, we focus on the works of Jin Yong, a renowned Chinese novelist, to explore the effectiveness of Latent Dirichlet Allocation (LDA) and Support Vector Machine (SVM) in modeling and classifying Chinese literary texts. By uniformly sampling 1,000 paragraphs from his novels and varying key parameters — such as the number of topics (T), the basic units (“word” vs. “character”), and paragraph length (K) — we aim to systematically evaluate the performance of LDA in capturing semantic structures and the impact on classification accuracy. Using 10-fold cross-validation, we assess the model's robustness and provide insights into optimal configurations for practical text analysis tasks. This work not only enhances the understanding of topic modeling in Chinese literature but also offers data-driven recommendations for model selection and optimization in similar contexts.

Methodology

1. Latent Dirichlet Allocation

Latent Dirichlet Allocation (LDA) constitutes a probabilistic generative model for text corpora that operates under the fundamental assumption that documents manifest as mixtures of latent topics, where each topic is characterized by a probability distribution over words. This hierarchical Bayesian framework employs Dirichlet priors to model document-topic (θ_d) and topic-word (φ_k) distributions, thereby establishing a complete data generation mechanism.

The generative process of LDA unfolds through the following probabilistic steps: For each document d in the corpus, the model first draws a topic proportion vector θ_d from a Dirichlet distribution with parameter α , expressed as $\theta_d \sim \text{Dir}(\alpha)$. Concurrently, for each topic k , it generates a word distribution φ_k from another Dirichlet distribution with parameter η , denoted as $\varphi_k \sim \text{Dir}(\eta)$. When generating the n th word in document d , the model initially samples a topic assignment $z_{d,n}$ from the document-specific topic distribution ($z_{d,n} \sim \text{Multinomial}(\theta_d)$),

subsequently drawing the observed word $w_{d,n}$ from the corresponding topic's word distribution ($w_{d,n} \sim \text{Multinomial}(\phi_{z_{d,n}})$).

The complete joint probability distribution of the model factors as:

$$P(W, Z, \theta, \phi | \alpha, \eta) = \prod_{d=1}^D P(\theta_d | \alpha) \prod_{k=1}^K P(\phi_k | \eta) \prod_{n=1}^{N_d} P(z_{d,n} | \theta_d) P(w_{d,n} | \phi_{z_{d,n}})$$

This factorization explicitly represents the hierarchical dependencies between parameters and observed variables. The Dirichlet hyperparameters α and η exert crucial control over the sparsity patterns in topic distributions, where smaller values of α encourage documents to concentrate on fewer topics, while reduced η values promote topics dominated by fewer characteristic words.

The generative process finds its formal representation through the plate notation: $\alpha \rightarrow \theta_d \rightarrow z_{d,n} \rightarrow w_{d,n} \leftarrow \phi_k \leftarrow \eta$, where α and η denote hyperparameters, θ_d and ϕ_k represent latent variables, $z_{d,n}$ corresponds to topic assignments, and $w_{d,n}$ signifies observed word tokens. This graphical model elegantly captures the conditional independence relationships inherent in the LDA framework.

As a fully generative model, LDA provides not only a principled approach to document modeling but also establishes the theoretical foundation for subsequent variational inference or Gibbs sampling procedures. The model's capacity to uncover latent semantic structures while maintaining computational tractability has rendered it particularly valuable for diverse applications including topic discovery, document classification, and information retrieval within the computational linguistics and text mining communities.

2. Support Vector Machine

The Support Vector Machine (SVM) is a supervised learning algorithm designed for classification and regression tasks. Its core principle is to identify an optimal hyperplane in a high-dimensional feature space that maximally separates data points of different classes. For linearly separable data, this hyperplane is derived by solving a convex optimization problem that maximizes the margin—the distance between the hyperplane and the nearest data points (support vectors) from each class.

The experiment in this study developed a modular text classification system combining a Latent Dirichlet Allocation (LDA) topic model and a Support Vector Machine (SVM) classifier. The implementation began by initializing global parameters, including paragraph length and a predefined list of topic numbers (T), alongside integrating memory monitoring tools to track computational resource utilization. A core cross-validation function was designed to encapsulate the entire workflow: LDA model training for topic distribution extraction, transformation of text segments into topic probability vectors, SVM classifier construction, and performance evaluation. Additionally, a dedicated function was implemented to systematically process each topic number configuration. The main execution strategy adopted a progressive approach: preliminary small-scale tests with reduced parameter sets (e.g., shorter paragraphs, fewer topics) were conducted to validate system stability and workflow feasibility, followed by parallelized large-scale experiments to evaluate performance across all predefined T values. For each topic number setting, a rigorous 10-fold cross-validation procedure was applied, generating the mean classification accuracy and standard deviation as key metrics. The system architecture emphasized modularity, robustness, and reproducibility, incorporating real-time resource monitoring to prevent memory overflow and consistency checks to ensure algorithmic reliability. This design not only enhanced debuggability during development but also facilitated scalable extensions for future text analysis tasks.

The experiment comprises three key steps: data preprocessing, model construction and training, and model evaluation.

Step 1: Data Preprocessing:

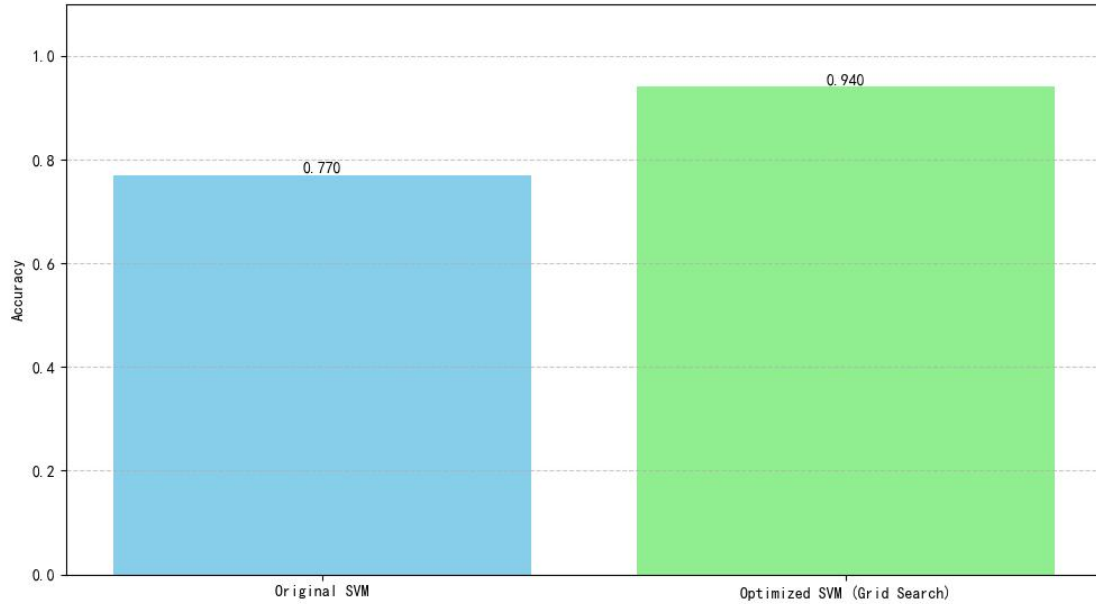
From a designated corpus, 1,000 paragraphs were uniformly sampled, and the text data from the txt files were read and stored as lists. For character-level experiments, punctuation and stopwords were first removed from the text data, after which each Chinese string in the list was segmented into individual Chinese characters and stored as a new list. For word-level experiments, punctuation was first eliminated, followed by Chinese word segmentation using the jieba library; subsequently, stopwords were removed, and the processed text was stored as a new list.

Step2: Model Construction and Training

The Latent Dirichlet Allocation (LDA) model was employed to generate topic distributions. The segmented text fragments were transformed into LDA topic distribution vectors, and new text fragments were converted into LDA features for prediction using a Support Vector Machine (SVM) classifier as the classification model. When the paragraph length (K) was fixed at 1,000, the number of topics (T) was set to 5, 10, 20, 40, and 80, respectively. Conversely, when the number of topics was fixed at 20, the paragraph lengths were set to 20, 100, 500, 1,000, and 3,000. During model training, grid search was utilized to determine optimal parameters, which were then used as

the initial parameters for the SVM model to enhance accuracy. Additionally, a progressive testing strategy was implemented—small K values (20, 100) were tested first to verify the workflow's feasibility, and the full parameter set was executed only after successful validation of these smaller values, thereby improving code debuggability.

Comparison of Model Accuracies: Before vs After Grid Search



Step 3: Model Evaluation:

A 10-fold cross-validation was conducted, and the average accuracy along with its standard deviation were computed to assess the classification performance of the model.

Experimental Studies

The experiments in this study include a controlled experiment and a full factorial design. First, a controlled variable approach is employed to systematically investigate the impact of different parameters on classification performance, with the experimental design encompassing three dimensions:

1. Topic Quantity Dimension:

With paragraph length fixed at $K = 1000$, the influence of topic numbers ($T \in \{5, 10, 20, 40, 80\}$) on model accuracy and stability is systematically evaluated under both word-based and character-based segmentation.

2. Paragraph Length Dimension:

With the topic number fixed at $T = 10$, classification performance is measured across varying paragraph lengths ($K \in \{20, 100, 500, 1000, 3000\}$) using 10-fold cross-validation (reported as mean accuracy $\pm$ standard deviation) for both basic units of words and characters.

3. Text Granularity Dimension:

Under identical configurations of paragraph length and topic number, the classification performance of basic units of words is compared with that of basic units of characters.

Performance Under Word-Based Segmentation (Fixed $T = 10$, Varying K)

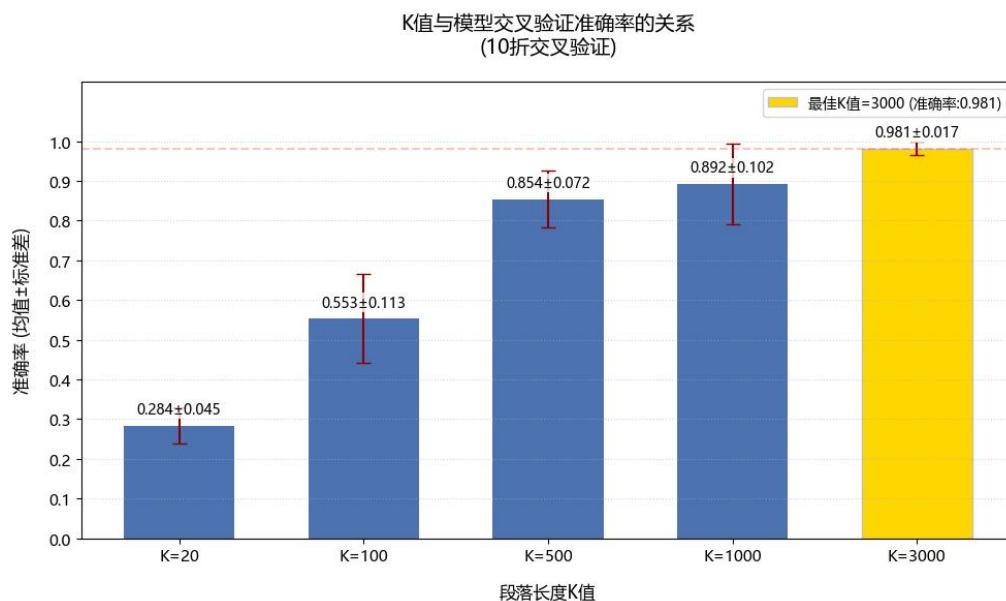


Figure 1 : The mean accuracy and standard deviation across varying values of K (word)

As illustrated in Figure 1, when the number of topics is 10, the classification performance significantly improves with longer paragraphs, such as K=3000, achieving an average accuracy rate of 0.981 ± 0.017 in 10-fold cross-validation, whereas the accuracy rate for short paragraphs, such as K=20, is only 0.284 ± 0.045 . This discrepancy may stem from the richer context information provided by longer paragraphs, which allows LDA to capture a more stable and coherent topic distribution, thereby providing the SVM classifier with more discriminative feature vectors. Additionally, the standard deviation significantly decreases with longer paragraphs (e.g., 0.017 for K=3000 and 0.102 for K=100), indicating that longer texts reduce the volatility in topic modeling due to data sparsity, thus enhancing the robustness of the model.

Performance Under Word-Based Segmentation (Fixed K = 1000, Varying T)

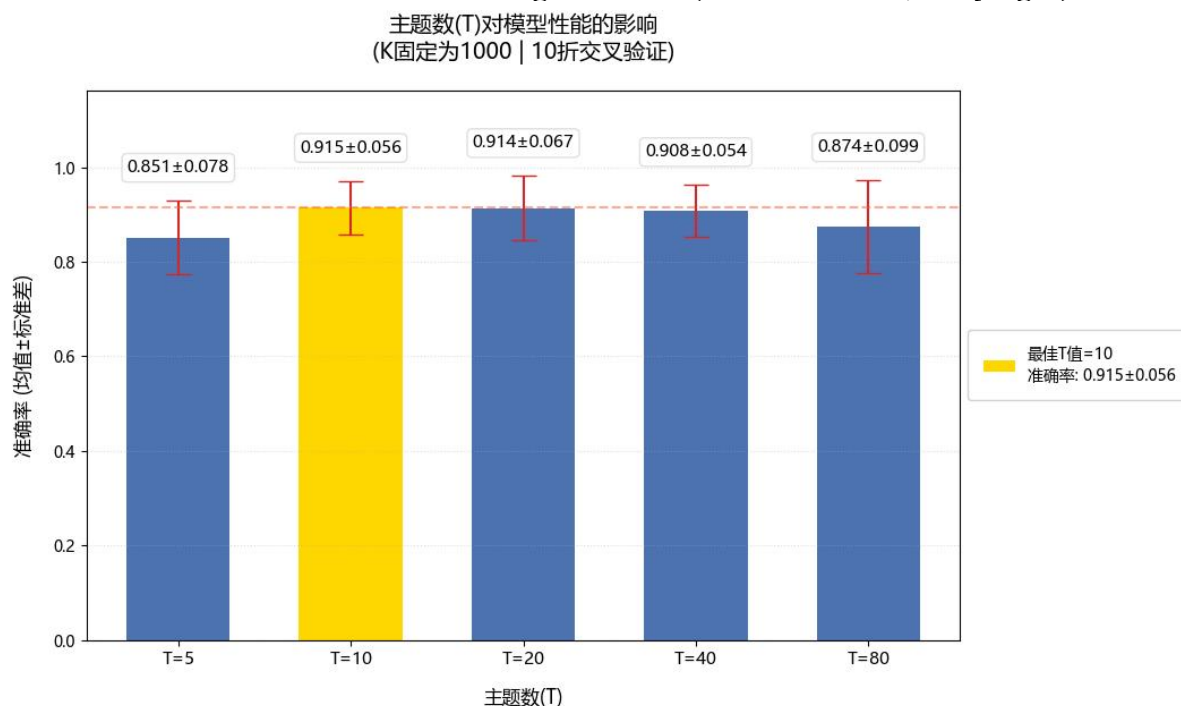


Figure 2 : The mean accuracy and standard deviation across varying values of T (word)

The results in Figure 2 demonstrate that the model's accuracy rate reaches its peak at 0.915 ± 0.056 when T=10, indicating that T=10 is the optimal choice of topic number in this specific experimental setup. The chart also displays the error range (standard deviation) for each topic number. For instance, the error range for T=10 is ± 0.056 , suggesting that the variation in accuracy rate across multiple experiments is relatively small, and the model's performance is relatively stable. Overall, the model's accuracy rate first increases and then decreases with the increase in the

number of topics. An insufficient number of topics may prevent the model from adequately capturing the complexity of the data, while an excessive number of topics may lead to overfitting, thus reducing the model's generalization ability on unseen data.

Performance Under Character-Based Segmentation (Fixed T = 10, Varying K)

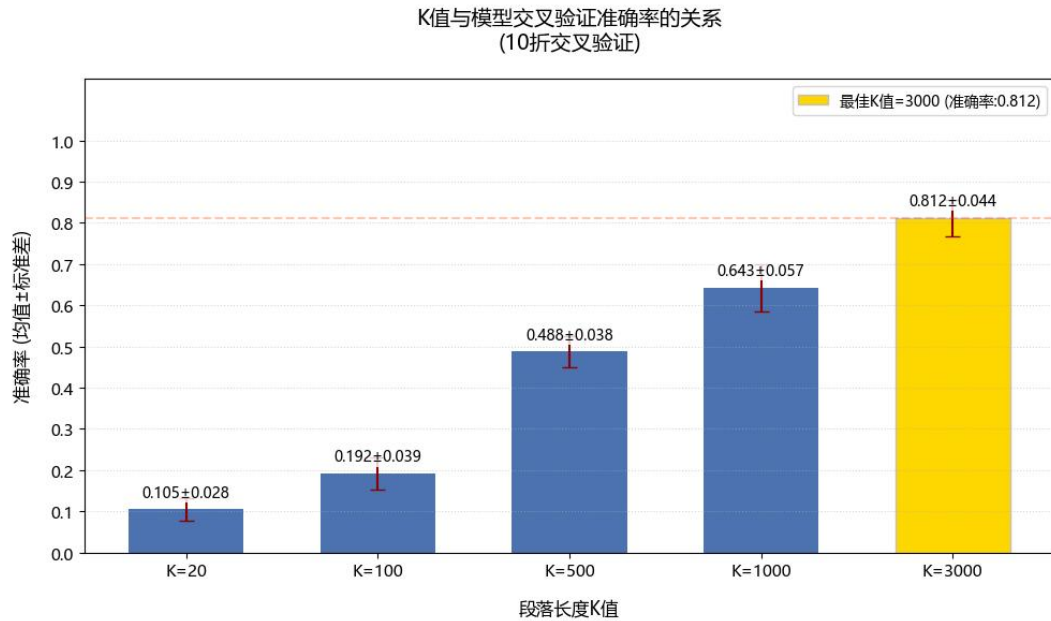


Figure 3 : The mean accuracy and standard deviation across varying values of K (character)

As shown in Figure 3, for basic units of character, the model's accuracy rate gradually improves with the increase in paragraph length (K). Shorter paragraph lengths may prevent the model from fully capturing the complexity of the data, while longer paragraphs may provide more context information, allowing the model to better learn and distinguish between subtle differences between different categories.

Performance Under Character-Based Segmentation (Fixed K = 1000, Varying T)

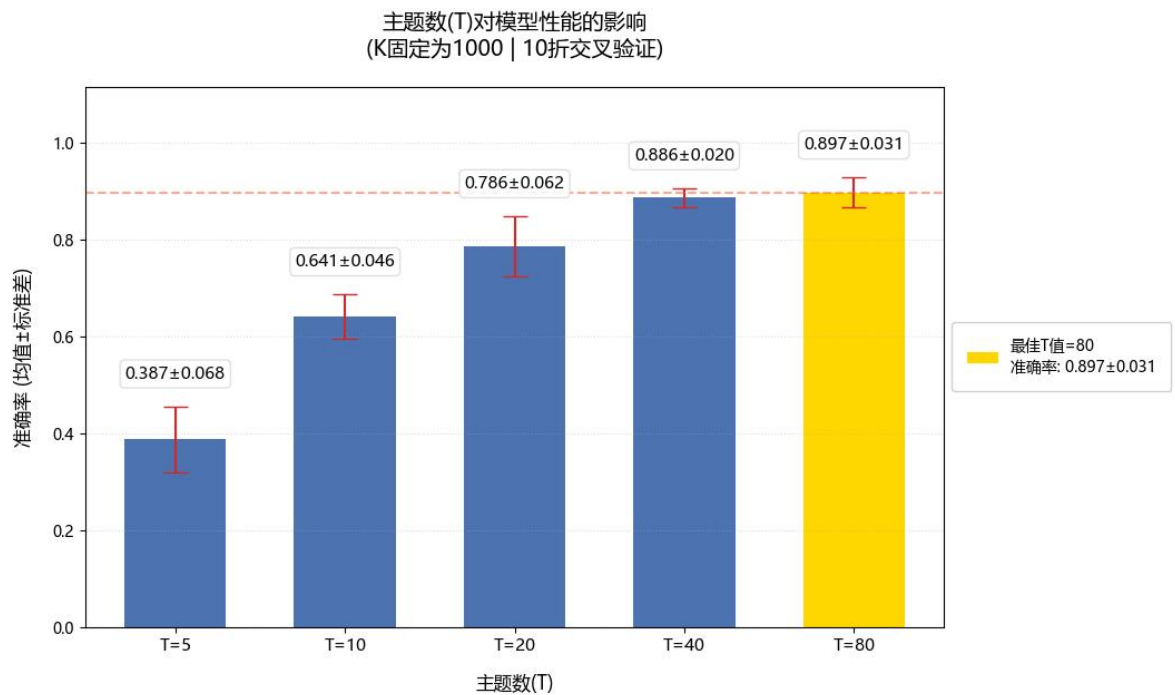


Figure 4 : The mean accuracy and standard deviation across varying values of T (character)

The figure 4 shows that with the increase in the number of topics (T), the model's accuracy rate generally trends upward. This indicates that a greater number of topics can help the model better capture the complexity and diversity of the data.

While the controlled variable approach allows for clear quantification of the independent effects of single variables on classification performance with low computational costs, it fails to capture synergistic or antagonistic interactions between variables. To address this limitation, we further implemented a full factorial experiment by simultaneously varying paragraph length ($K \in \{20, 100, 500, 1000, 3000\}$ tokens) and topic quantity ($T \in \{5, 15, 30, 60, 80\}$), resulting in 25 experimental configurations. This design enables systematic analysis of the combined effects of parameter combinations on classification performance. For data visualization, the experimental results were visualized through heatmaps and complemented by three-dimensional surface plots. These surface plots exhibit a smoothed 3D surface, where classification accuracy varies as functions of paragraph length (K) and topic quantity (T). The surface color gradient transitions from light hues (low accuracy) to dark hues (high accuracy), with raw data points overlaid in red on the surface to facilitate validation of interpolation fidelity.

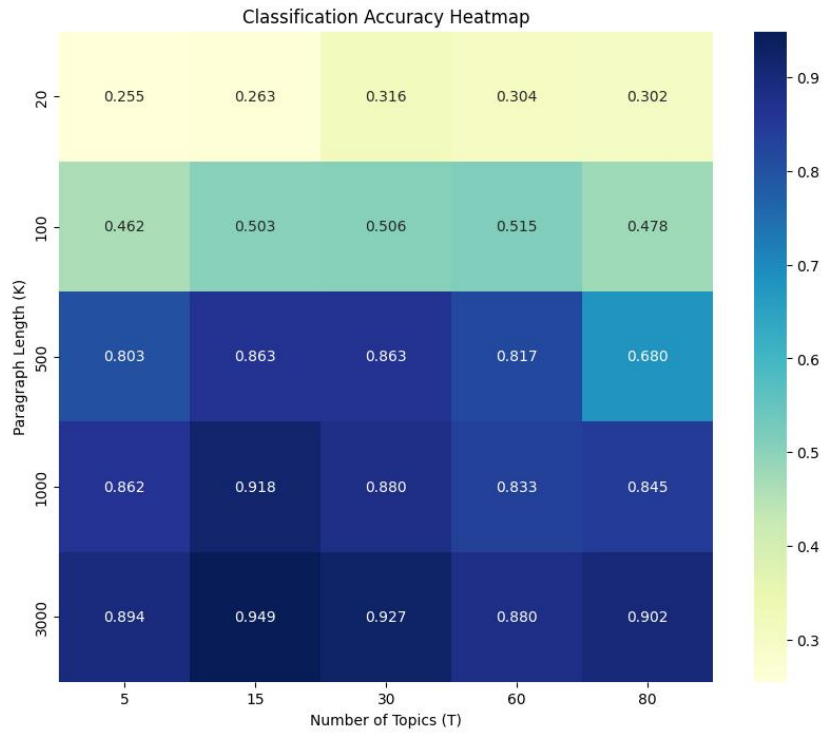


Figure 5 : The mean accuracy across varying values of T and K (word)

As shown in Figure 5, when using words as the basic unit, the mean classification accuracy reaches its peak value at $T=15$ and $K=3000$, indicating the existence of an optimal range for topic quantity. Classification performance is significantly positively correlated with paragraph length. Shorter paragraphs (e.g., $K=20$ and 1000) exhibit lower classification accuracy, but demonstrate notable improvement as T increases. In contrast, longer paragraphs (e.g., $K=1000$ and 3000) achieve high classification accuracy even with smaller T , with limited scope for further improvement as T increases. This suggests that longer texts provide richer contextual information, enabling effective classification with fewer topics.

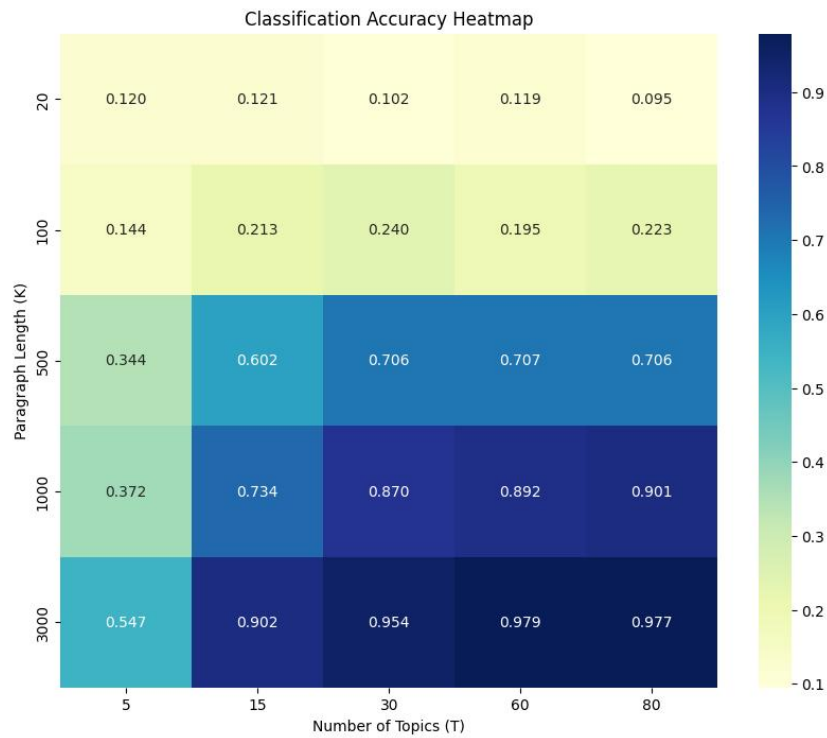


Figure 6 : The mean accuracy across varying values of T and K (character)

As depicted in Figure 6, when employing characters as the fundamental units, the classification accuracy achieves its peak at $T=60$ and $K=3000$. Furthermore, there exists an optimal range for the number of topics. A significant positive correlation is observed between classification performance and paragraph length. With the increment in the number of topics (T), there is a general enhancement in classification accuracy. For $K=20$, the classification accuracy is remarkably low, nearly approaching the level of random guessing, which can be attributed to the insufficient information provided by short texts, thereby impeding effective differentiation among various categories. When $K=100$ or 500 , a notable increase in classification accuracy is observed with the rise in T . For $K=1000$ and 3000 , even at lower values of T , the classification accuracy remains high. Although further increases in T lead to enhanced accuracy, the rate of improvement is not as pronounced as that observed in shorter texts.

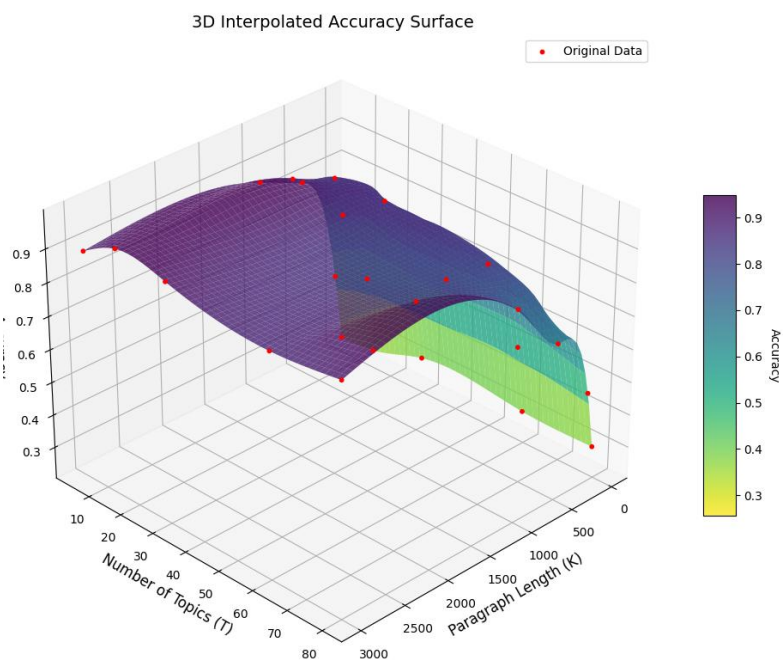


Figure 7 : The 3D interpolated accuracy surface across varying values of T and K (word)

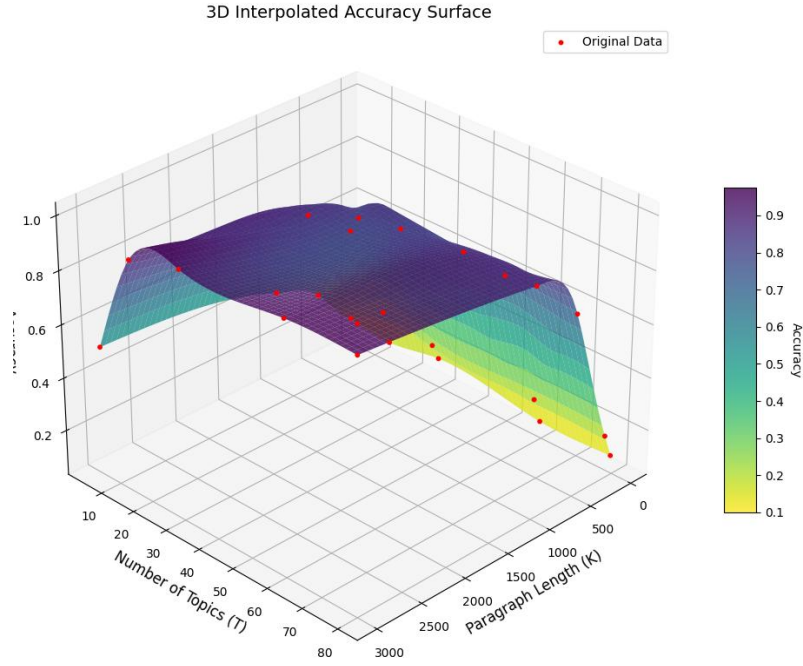


Figure 8 : The 3D interpolated accuracy surface across varying values of T and K (character)

According to Figure 7 and Figure 8, as evidenced by the trivariate interpolation surface plot utilizing words and characters as basic textual units, the relationship between parameters K, T, and the model's predictive accuracy is nonlinear.

Conclusions

This study systematically investigated the independent effects and interactions of paragraph length, number of topics, and token units on text classification performance through controlled experiments. The results demonstrated a significant positive correlation between paragraph length (K) and classification accuracy. Longer paragraphs (e.g., $K=3000$) provided richer contextual information, enabling the LDA model to capture more stable and consistent topic distributions, thereby offering more discriminative feature vectors for the SVM classifier. Long paragraphs ($K \geq 1000$) achieved high classification accuracy even with a smaller number of topics (T), indicating that long texts effectively alleviated the fluctuation in topic modeling caused by data sparsity, enhancing model robustness. In contrast, short paragraphs (e.g., $K=20$) suffered from insufficient information, resulting in classification accuracy close to random guessing, although their performance could be partially improved by increasing the number of topics (T).

The number of topics (T) exhibited a clear optimal range for model performance. Peak classification accuracy was achieved at $T=10$ for word units and $T=60$ for character units, respectively (0.915 ± 0.056 and peak accuracy), suggesting that a moderate number of topics effectively balanced data complexity and model generalization capability. Too few topics failed to adequately represent the semantic structure of texts, while too many topics could lead to overfitting, reducing the model's adaptability to new data. Additionally, the impact of the number of topics was more pronounced for short paragraphs ($K \leq 500$), with accuracy significantly improving as T increased; for long paragraphs ($K \geq 1000$), the marginal benefit of T diminished, indicating that the inherent information redundancy in long texts supported efficient classification even with a lower number of topics.

The experiments also revealed differences in classification performance between word units and character units. Word units had a significant advantage in semantic expression, particularly in long text ($K=3000$) scenarios, where their peak accuracy exceeded that of character units. This difference arose because word units could directly capture lexical-level semantic features, whereas character units relied on more complex combination patterns. However, character units showed greater robustness in handling out-of-vocabulary words and had greater potential for improving the classification performance of short texts ($K \leq 100$). A common pattern for both units was that performance tended to stabilize for long texts ($K \geq 1000$) and was highly dependent on the adjustment of the number of topics for short texts ($K \leq 100$).

A three-dimensional interpolation surface plot illustrated the nonlinear interaction between paragraph length (K), number of topics (T), and classification accuracy. For instance, short texts ($K=20$) required a high number of topics ($T \geq 60$) to compensate for information loss, while long texts ($K=3000$) achieved optimal performance with a moderate number of topics ($T=15 \sim 30$). This nonlinear relationship suggests that the number of topics should be dynamically optimized based on text length in practical applications.

The experimental design of this study was based on a fixed corpus of novels, and future work could extend to multi-domain texts (such as news, social media) to validate the generalizability of the conclusions. Additionally, the limited representation capacity of topic models (LDA) for short texts warrants exploration into combining pre-trained language models (such as BERT) or dynamic topic modeling (DTM) to enhance short text classification performance. Finally, the parameter optimization process could be further accelerated through Bayesian optimization or automated machine learning (AutoML) frameworks to support large-scale application scenarios.